

Course Machine Learning

Course #1 – Introduction to ML

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1 - Introduction

In this section we will introduce the basic principles of machine learning, the different concepts and definitions related to this topical field as well as the different tasks and functions of Machine Learning.

Definition of Machine Learning

Machine **learning** is a field of **AI** that aims to **give machines the ability to learn**. This powerful technology has enabled the development of **self-driving cars**, **voice recognition**, and all so-called "**intelligent**" systems over the past 10 years.

Machine learning was invented by **Arthur Samuel** in 1959, after he developed the first **checkers program** with artificial intelligence. This program taught itself to play checkers without receiving any instructions from its developer. "*How did he do it?*" We will see this

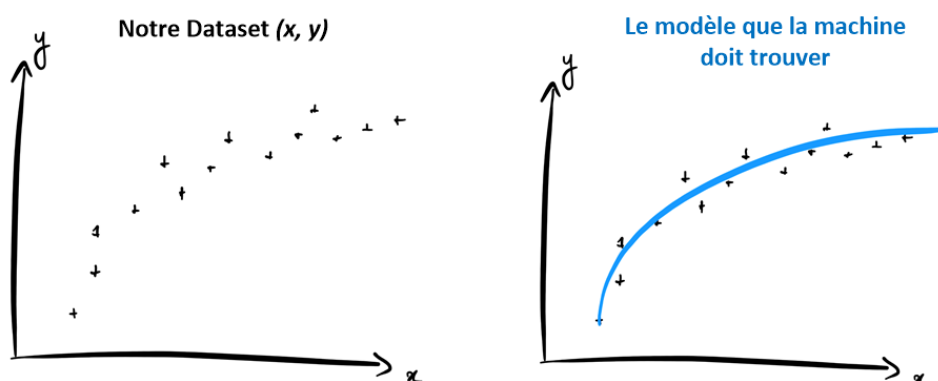
later in this article.

Following his invention, *Arthur Samuel* formulated the historic **definition** of Machine Learning:

" **Machine learning** is the science of giving a machine the **ability to learn**, without explicitly programming it. "

Statistical Learning

Today, we have a more mathematical way of defining Machine Learning. In French, we call it **Statistical Learning** (or automatic learning) because the majority of algorithms used in Machine Learning are actually **statistical models** developed **from data**. Examples of these models include *decision trees*, *linear regression*, and *Bayesian models*.



Machine learning involves letting the machine develop a **statistical model** from **data** provided to it.

So, for a machine to learn to recognize a face in a photo, we provide it with **thousands of photos** to study, and the machine develops its own **statistical rules** to differentiate between faces.

How does Machine Learning work?

To understand how a machine can learn from data, we need to look at the 3 learning paradigms of Machine Learning:

- Supervised **Learning**
- **Unsupervised** Learning
- Reinforcement **Learning**

2 — Supervised Learning: Definition

Supervised Learning is used to develop **predictive models**, i.e. models capable of predicting a certain value y based on variables x_1, x_2 , etc.

Examples: Predicting the name of a person (y) from a photo (x), or predicting the price of an apartment (y) based on its living area (x_1) and the number of rooms (x_2)

To develop such models, the machine must first be provided with a large amount of **data** (x, y). This is called a **dataset**. Then, the machine is asked to develop an **approximation function** that best represents the relationship $x \rightarrow y$ present in our data. For this, we use an **optimization algorithm** that **minimizes the discrepancies** between the function and the dataset data.

Note: The name "Supervised Learning" comes from the fact that we supervise (like a teacher) the machine's learning by providing it with examples of **questions/answers**. (x/y) to study.

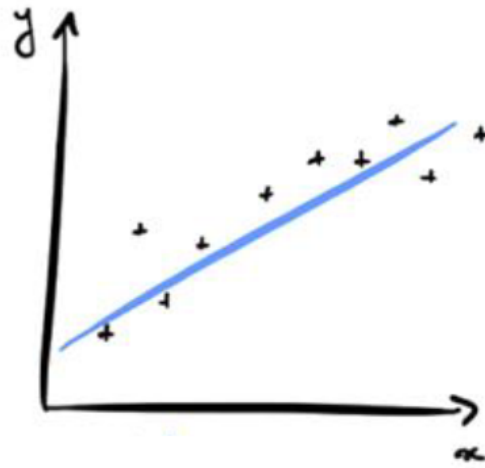
The **applications** of Supervised Learning are numerous, very numerous! They can be divided into two categories of problems: **Regressions** and **Classifications**.

Regression Problems

Regression problems correspond to situations in which the machine must predict the value of a **quantitative variable** (continuous variable)

Examples of y variables :

- the price of an apartment
- climate change
- the stock market price
- patient lifespan



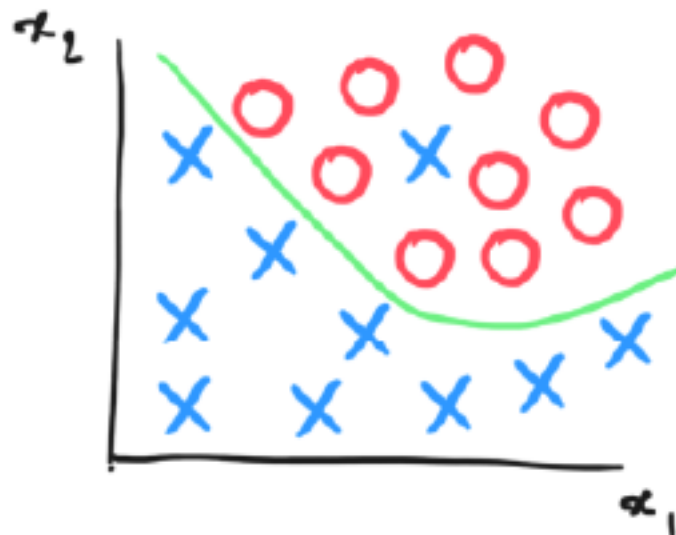
Representation of a regression problem

Classification problems

Classification problems correspond to situations in which the machine must predict the value of a **qualitative variable** (discrete variable). In other words, the machine must **classify** what is given to it into classes.

Examples of y variables :

- Email Spam / Non Spam
- Cancer / non-cancer
- Photo of Cat / Dog



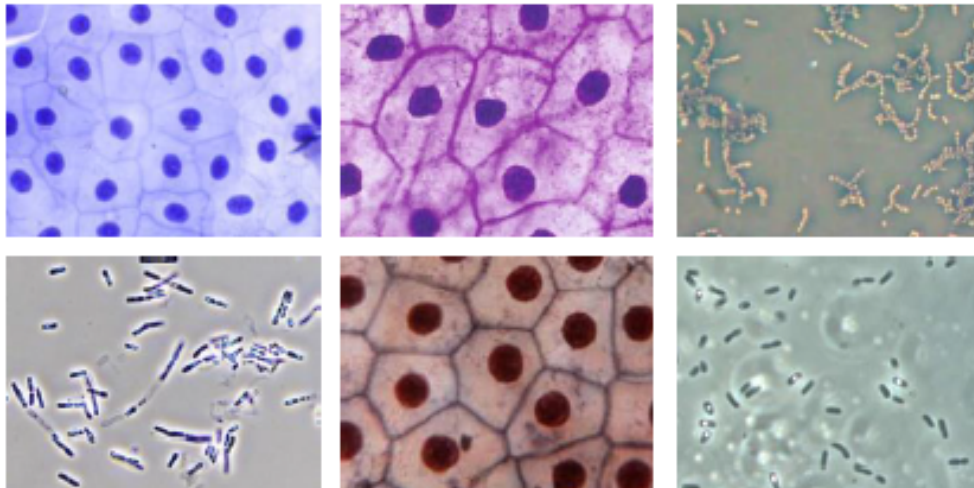
Representation of a classification problem

Note: Speech recognition and computer vision are classification problems. The machine learns to classify the sound/image signals it receives into different word packets.

2 — Unsupervised Learning: no variable y

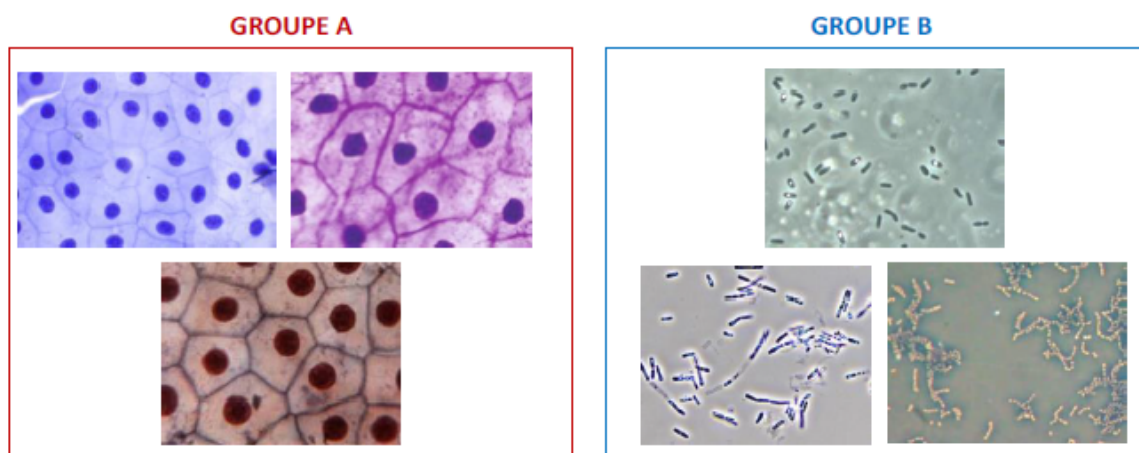
Another learning method for developing Machine Learning programs is **machine learning. Unsupervised Learning. This method is** used when our dataset does not contain examples that indicate what we are looking for. Wait... let me explain with an analogy !

Look at these 6 photos. Can you group them into 2 families based on their similarity?



Of course! It's actually quite simple. You don't need to know whether they're animal cells, bacteria, or proteins to learn how to classify these images . Your brain has actually recognized **common patterns** in the data you've shown it.

In unsupervised learning, we have a Dataset x without variable y , and the machine **learns** to recognize **structures** in the data x that we show it.



We can thus group data into clusters (this is **Clustering**), detect **anomalies** , or even **reduce the** very rich data **dimension by compiling the dimensions together.**

3 — Reinforcement Learning: learning in one's environment

A final very popular technique is **learning by reinforcement** which consists of letting the machine learn to do a task (for example piloting a mini drone) by letting it practice on its own. When the machine succeeds in what it undertakes, it receives a **bonus** . When it fails, it receives a **penalty** .

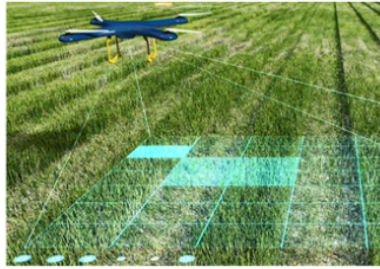
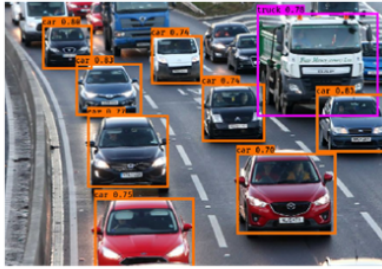
We develop a program that forces the machine to want **to maximize** its bonuses, and the machine then analyzes its own past mistakes in order to **improve** over time. It's a bit like when we learn to ride a bike: at first we don't succeed at it at all, but by doing it, we intrinsically develop our balance in order to avoid the past mistakes that made us fall!



Conclusion: Machine Learning is the new electricity, it will revolutionize everything.

In 2019, machine learning is already all around us. In fact, you probably use it **hundreds of times a day** without even realizing it. Every time you search Google, a machine learning model has learned how to rank the most relevant results on the first page of millions of possible web pages.

When you post a photo of yourself on Facebook, there is a machine learning algorithm that manages to identify you because it has learned to recognize faces in photos.



Machine learning has already begun to change the face of our world. It's revolutionizing the transportation industry with self-driving cars and powering our connected devices: iPhones, voice recognition, and computer vision. It diagnoses cancer better than a team of doctors and improves banking and IT security, makes fairer legal decisions, and more. Even the agricultural industry and the art world are affected by machine learning.

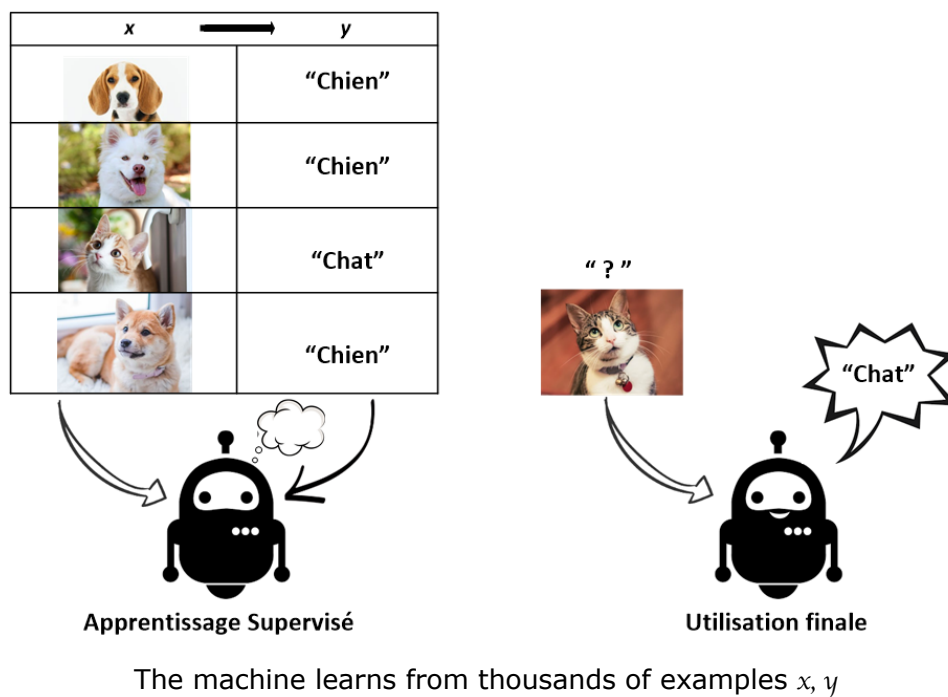
4 — Supervised Learning

Supervised **learning is the most popular** learning paradigm in Machine Learning and Deep Learning. As the name suggests, it involves supervising the machine's learning by showing it **examples (data) of the task** it must perform. There are many applications: speech recognition, computer vision, regressions, classifications, etc. The vast majority of Machine Learning and Deep Learning problems use **machine learning, supervised**. It is therefore essential to fully understand how this mechanism works.

How does supervised learning work?

With supervised learning, a machine can learn to perform a specific task by studying **examples** of that task. For example, it can learn to recognize a picture of a dog after being shown millions of pictures of dogs. Or, it can learn to translate French into Chinese after seeing millions of examples of French-Chinese translations.

Generally speaking, the machine can learn a **relationship** $f: x \rightarrow y$ which connects x to y by **analyzing** millions of examples of $x \rightarrow y$ associations.



Supervised learning works in 4 steps:

1. Import a **Dataset** (x, y) which contains our examples
2. Developing a **Model** with Random Parameters
3. Develop a **Function Cost** that measures the errors between the model and the Dataset
4. Develop a **learning algorithm** to find the model **parameters that minimize the function Cost**

1. The Dataset: Examples of what to learn

Data Science begins with the word... **Data** !

The first step of a Supervised Learning algorithm is therefore to import a Dataset which contains the examples that the machine must study.

This Dataset always includes 2 types of variables:

- A **target** variable y
- One or more characteristic variables (**features**) x

For example, imagine you are visiting a series of apartments on a Saturday afternoon. For each apartment, you enter in an Excel spreadsheet the price y and the characteristics x of the apartment (surface area, quality, city, etc.)

Exemple de Dataset sur des appartements

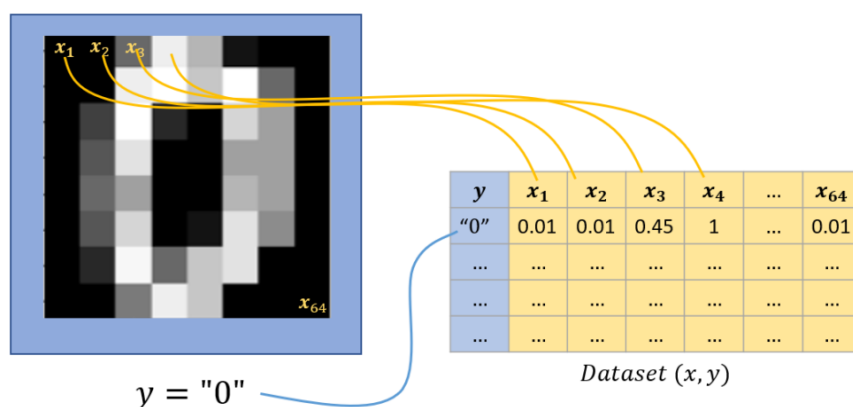
Target y		Features		
		x_1	x_2	x_3
Prix		Surface	Qualité	Adresse postale
313,000		90	3	95000
720,000		110	5	93000
250,000		40	4	44500
290,000		60	3	67000
190,000		50	3	59300
...		...	...	...

Par convention:
 m : nombre d'exemples
 n : nombre de features

By convention, we say that our Dataset contains m **examples** , that is, m rows. If you visited 6 apartments, then $m = 6$.

By convention, we also note n the number of **features** in our Dataset, that is to say the number of columns (except the y column). If you noted 3 characteristics for your apartment (Surface, quality, city), then $n = 3$.

When developing a computer vision program, the *features* of our dataset can be the pixels present in the image. A dataset of 8×8 pixels therefore gives 64 *features* ($n=64$). Each of these *features* will correspond to the value of the pixel (a black pixel = 1, a white pixel = 0)



Last convention: to designate a cell in our table, we generally note $x_{\text{colonne}}^{(\text{ligne})}$, that is to say that to designate the quality of the 3rd apartment that you visited, we write $x_2^{(3)}$

Dataset (x, y)

y	x_1	x_2	x_3	...	x_n
$y^{(1)}$	$x_1^{(1)}$	$x_2^{(1)}$	$x_3^{(1)}$	...	$x_n^{(1)}$
$y^{(2)}$	$x_1^{(2)}$	$x_2^{(2)}$	$x_3^{(2)}$	...	$x_n^{(2)}$
$y^{(3)}$	$x_1^{(3)}$	$x_2^{(3)}$	$x_3^{(3)}$	...	$x_n^{(3)}$
...	...	...	...	...	...
$y^{(m)}$	$x_1^{(m)}$	$x_2^{(m)}$	$x_3^{(m)}$	...	$x_n^{(m)}$

With such a Dataset, it becomes possible to predict new values y from values of x by developing a **model**, and this is our 2nd^{step} in solving a Supervised Learning problem!

2. The model: the heart of your program

The model is in some ways the heart of your program; it is what will perform the task you are trying to accomplish, for example, recognizing an animal in a photo or predicting the price of an apartment.

What is a model?

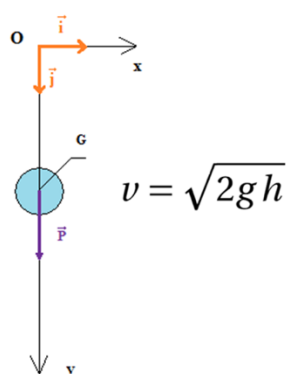
As a reminder, a model is a simplified **representation** of reality that can be used to **predict** what would happen under certain conditions. It can be a drawing, a physical equation, a mathematical function, a curve... in short, any representation.

For example, if I drop a 100g apple from a height of 4 meters, how long will it take for it to fall on good old Newton's head? We can **predict** this using Newton's **equations**. In this case, our model is **deterministic**: it will always give the same answer under the same conditions.

Réalité



Modèle



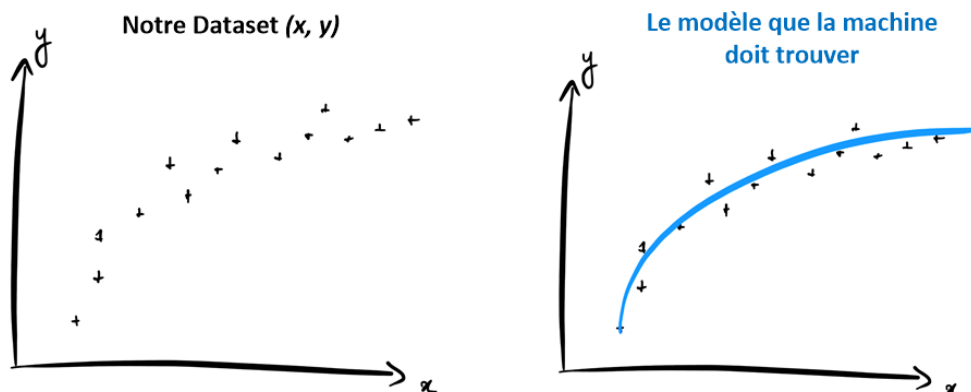
But to predict the price of an apartment based on all its characteristics, what is the mathematical equation to enter into the machine? And what is the equation to recognize a cat in a photo? No one knows!

The solution? **Leave there machine find** the model that best fits your **Dataset** (x, y) : this is **learning supervised**.

Machine Learning Models

Unlike the model illustrated above, a machine learning model is not based on a mathematical proof or a physical equation. Instead, it is built from data, like a statistical model.

If for example your Dataset gives you the following point cloud, then the machine will have to find the model that best fits this point cloud.

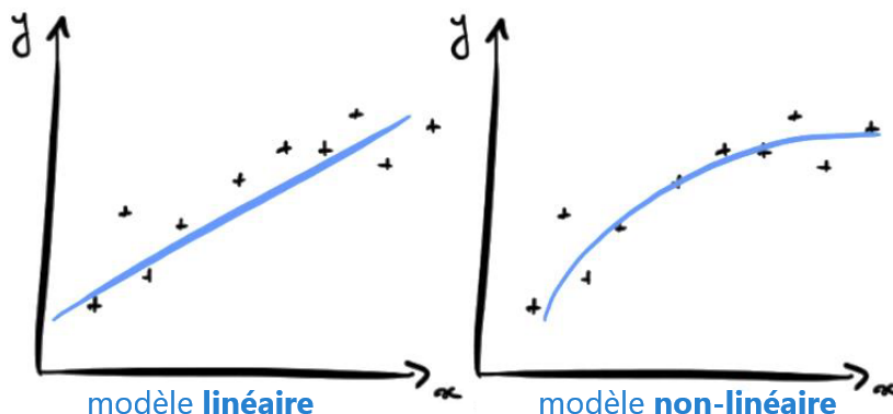


However, it is not up to the machine to do all the work!

In practice, it is up to us to choose the type of model (i.e. the mathematical function) and it is up to the machine to find the **coefficients** of this function that give the best results. By convention, we call these coefficients the **parameters** of the model.

For example, one can choose to develop a **linear model** $f(x) = ax + b$ and let the machine find the value of a and b that gives the best results.

Or we can choose a **non-linear model**, for example $f(x) = ax^2 + bx + c$, where a, b, c are the parameters. The possibilities are endless, but we will see later in this training how to choose one model over another.



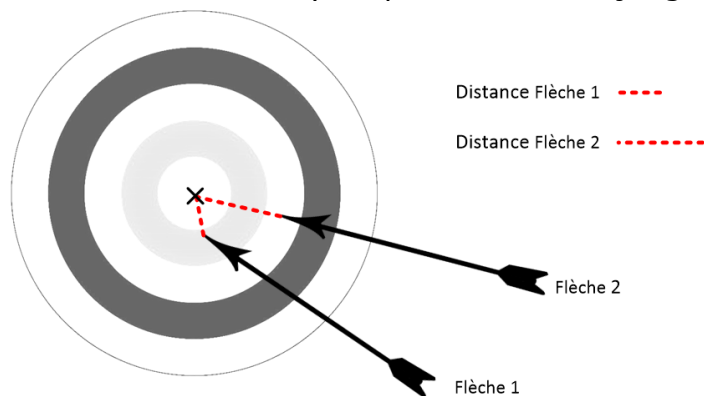
Now the question is: how do we make the machine find the best model? In other words, how do we make the machine learn?

The answer is in steps 3 and 4!

3. The Cost Function: Performance Measurement

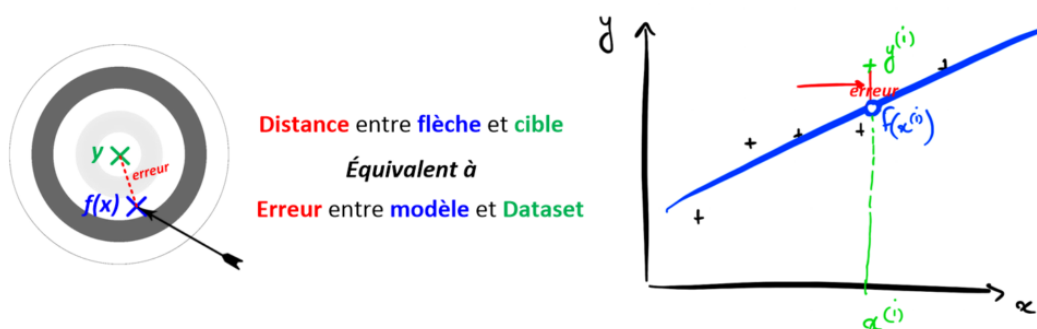
For the machine to find the best model, it must first be able to **measure there performance** of a given model. Don't believe me?

Imagine you're competing in an archery competition. How can you tell if you're better than your neighbor without measuring the **distance** between your arrows and the center of the target? It's impossible. You **MUST** measure your performance to judge which of you is better.

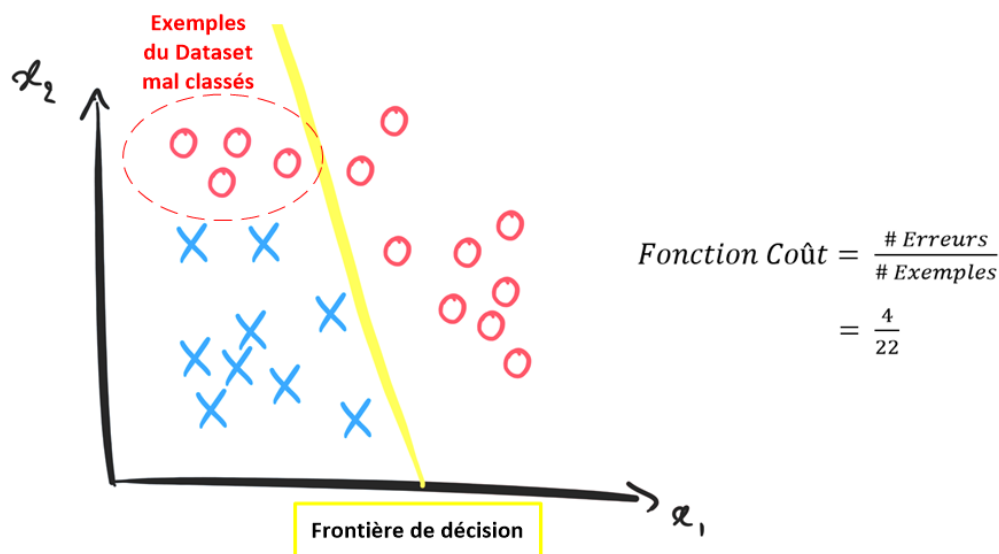


It's the same in Machine Learning. To know which model is the best among 2 candidates, we need **to evaluate them**. To do this, we measure the error between a model and the Dataset, and we call this the **Function Cost**.

In the case of **regression**, we can for example measure the error between the model prediction $f(x^{(i)})$ and the value $y^{(i)}$ that is associated with this $x^{(i)}$ in our Dataset. This is similar to the idea of measuring the distance between your arrow ($f(x^{(i)})$) and the center of the target, which is none other than the point ($y^{(i)}$) that it is supposed to reach.



Now, in the case of a **classification**, we can construct our Cost Function by measuring the number of examples in the Dataset that our model will have misclassified with its **boundary of decision**



There are many metrics to measure our errors (MSE, MAE, RMSE, etc.) and we will already see one of them in the next article of this training, which will allow you to move on to concrete matters with this story of Cost Function.

So that's step 3 of our supervised learning problem. At this point we have:

1. A **Dataset**
2. A **Model**
3. A **Function Cost** that measures the errors between the model and the Dataset

It's time to move on to the most important step in Machine Learning, the one that brings our program to life: the learning step.

4. The Learning Algorithm

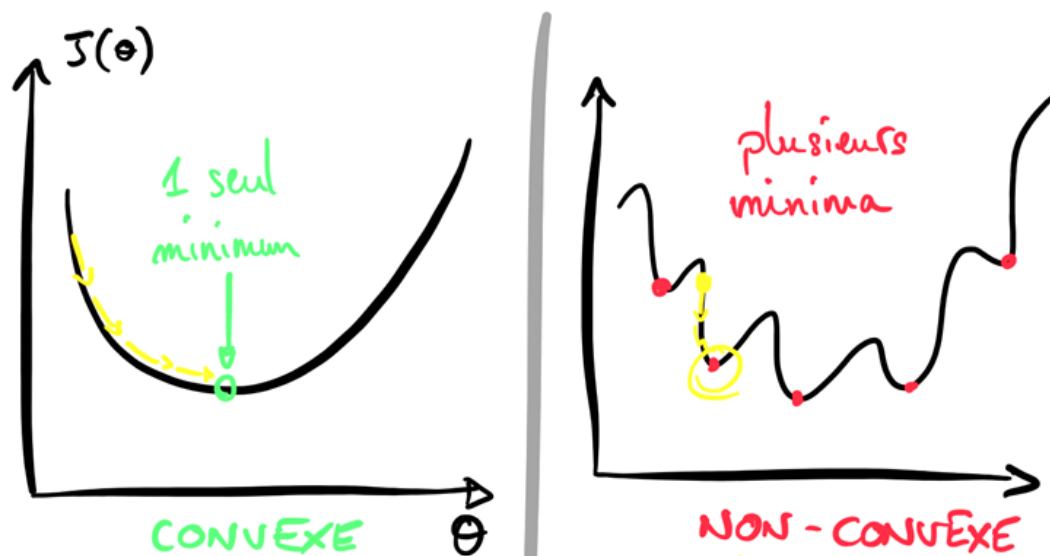
Let's talk little, let's talk well. Having a good model means having a model that has small errors. Logical?

So, in Supervised Learning, the machine looks for the model **parameters that minimize the Function Cost**. That's what we call learning. This sentence is **very important**. This is the essence of what you need to understand in Machine Learning.

To find the parameters that minimize the Cost function, there are a bunch of strategies.

For example, we could develop an algorithm that randomly tries several parameter combinations and chooses the combination with the lowest Cost Function. This is a bit like organizing an archery competition to keep only the best. However, this strategy is quite ineffective most of the time.

Another strategy, **very popular** in Machine Learning, is to consider the Cost Function as a **convex function**, that is, a function that has only one minimum, and to seek this minimum with a minimization algorithm called **Gradient Descent**. We will see this in detail in future articles. This strategy learns gradually, and ensures convergence towards the minimum of the Cost function (if convex).



In jargon, we call this step the **phase model training**. The machine chooses parameters for the model, then evaluates its performance (Cost Function) then looks for parameters that can improve its current performance, etc.

In Conclusion

You now know the 4 steps that give life to a supervised learning program. The mechanics are simple and applicable to any supervised learning problem with a Dataset (x, y) :

1. Import the **Dataset** and split it into a vector y and a matrix X
2. Create a **Model** f which produces a prediction based on certain conditions
3. Create a **Function Cost** that will be used to calculate the performance of the model on the examples in the Dataset.
4. Use an **Algorithm Learning** to find the model that minimizes the Cost Function.

This approach works just as well for regression problems (predicting the stock market price, predicting the travel time of a taxi, etc.) as for classification problems (detecting a cancer cell, sorting spam emails, etc.) and even for classifications in Deep Learning (computer vision, voice recognition, etc.)